ARJUNA

v2.1 · Adaptive Risk & Regime Allocation System

AQR / Bridgewater-style risk investing — for retail Indian equities

2.02	11%	0.997	0.01
Sharpe	max DD	Defl. Sharpe	PBO

15 stocks · quarterly · HRP · regime · Global Risk · sector≤2

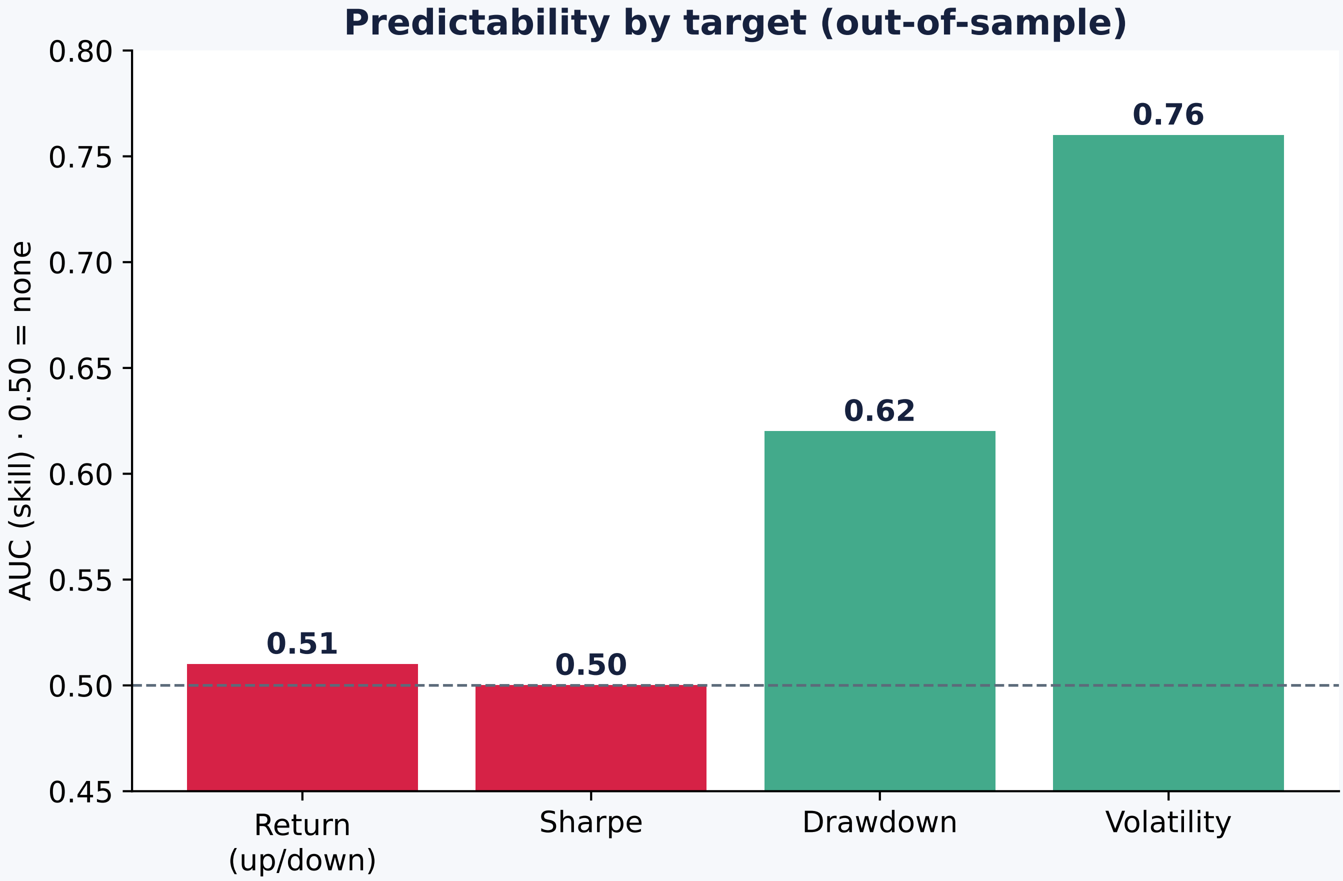
NOT a stock-picker. A risk-allocation & regime-management system.

Validated on clean Angel One data · Nifty-200 · net of cost · ~5.5y
Returns are unpredictable. Risk is. ARJUNA forecasts risk, not winners.

THE CORE PRINCIPLE

Returns are noise. Risk has structure.

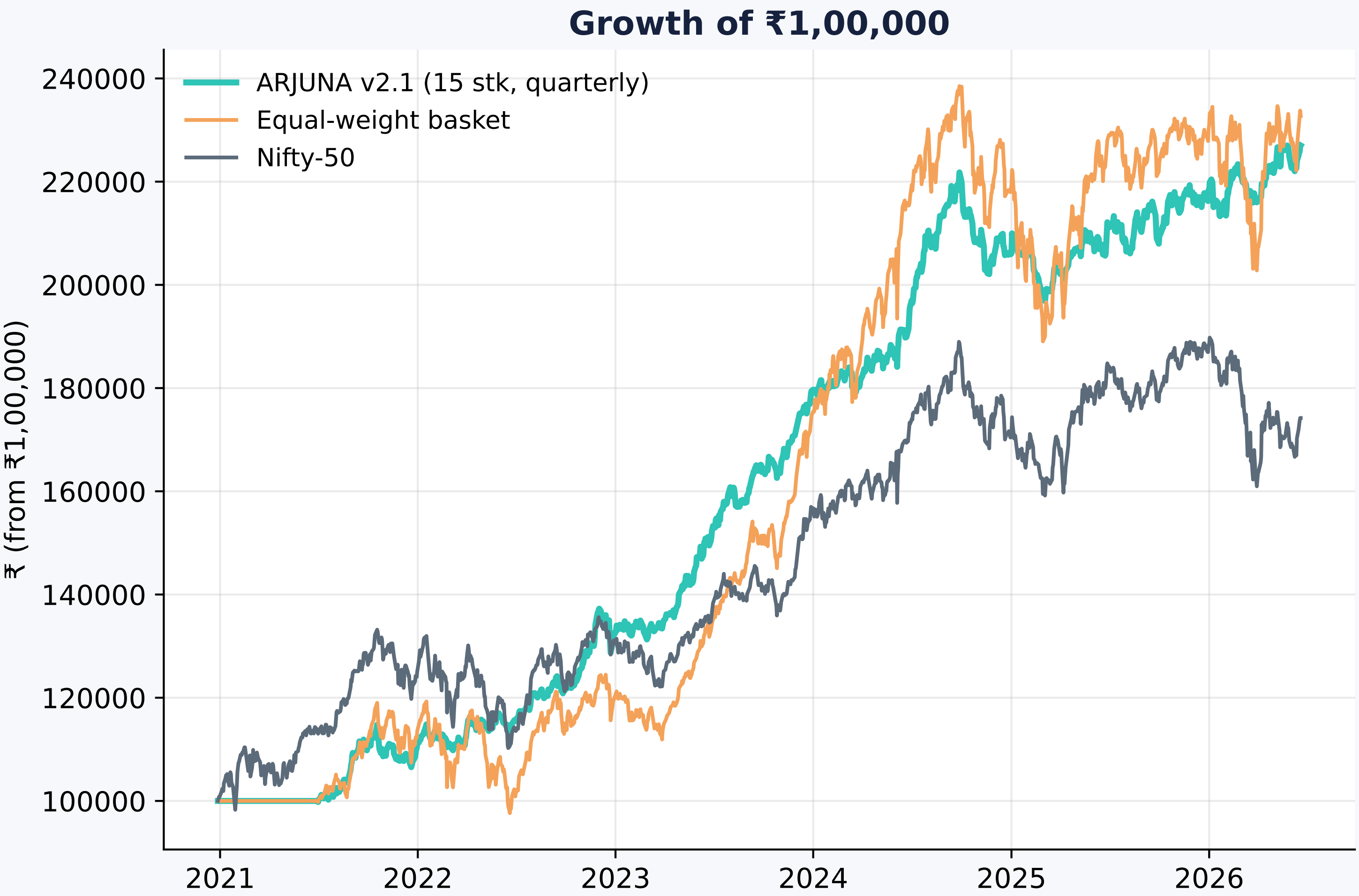
13 model families predicting RETURNS all scored ~0.50 (coin flip).
Change the TARGET to RISK and the same model gains real skill.



Forecast RISK · REGIME · EXPOSURE · WEIGHTS — never returns.
Objective: long-term risk-adjusted compounding. Survival first; raw CAGR last.

VALIDATED RESULTS

v2.1 vs the index (₹1L, ~5.5y, net of cost)



16.4%	**2.02**	**11.2%**	**3.06**
CAGR	Sharpe	max DD	Sortino

Rigour gate
Deflated Sharpe 0.997 · PBO 0.01 ✓

vs Nifty
Sharpe 2.0 vs 0.80 · DD 11% vs 17%

Survivorship inflates absolute CAGR -> trust the Sharpe/DD EDGE over the index.

ARCHITECTURE

Risk in, portfolio out

LAYER 1 · MARKET STATE

Global Risk (S&P/VIX/USD) · regime (VIX+200DMA) · breadth · FII/DII

LAYER 2 · RISK FORECAST

Volatility (AUC 0.76) · drawdown · trailing covariance

LAYER 3 · CORRELATION

Ledoit-Wolf cov · HRP clusters · sector≤2 (risk control)

LAYER 4 · PORTFOLIO

HRP weights · 15 stocks · QUARTERLY · news blow-up filter

FINAL PORTFOLIO
15 stocks · quarterly · capital-aware

AI used for risk, regime & news — NOT 'which stock doubles'.

THE DISCIPLINE

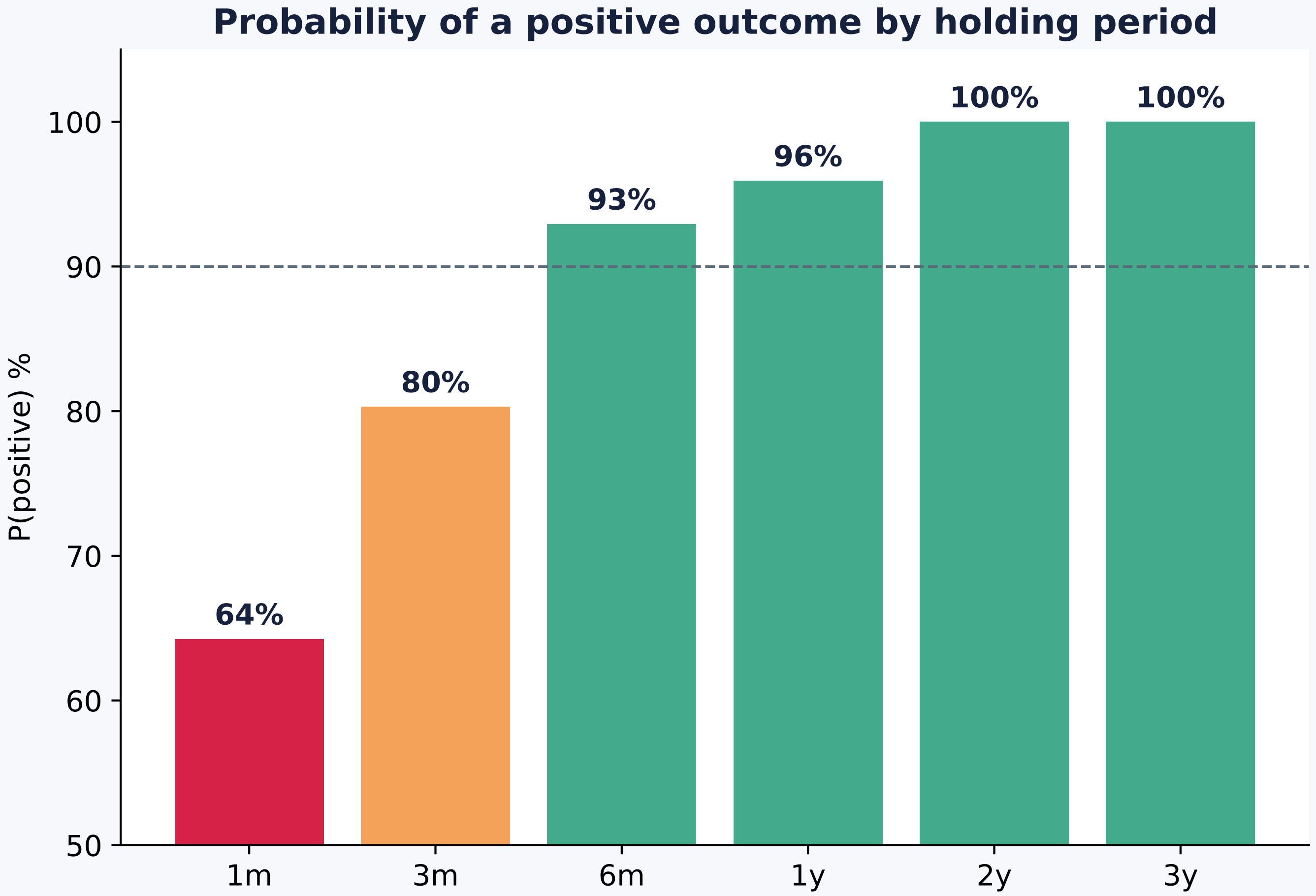
Evidence killed hypotheses — only survivors ship

SURVIVED	REJECTED (tested, failed)
Global Risk Engine	13 ML return-models (~0.50)
Regime (VIX+200DMA)	HMM regime / RL / GNN
HRP risk weighting	Sector-strength tilt (0.68)
QUARTERLY rebalance	Dynamic-N (0.97)
15-stock concentration	Exposure tiers (1.14)
Sector≤2 (diversify)	GARCH / vol-target / crash
News blow-up filter	Ranking / triple-barrier

Every rejection made ARJUNA simpler. Data is the bottleneck, not models.

HOW LONG TO HOLD

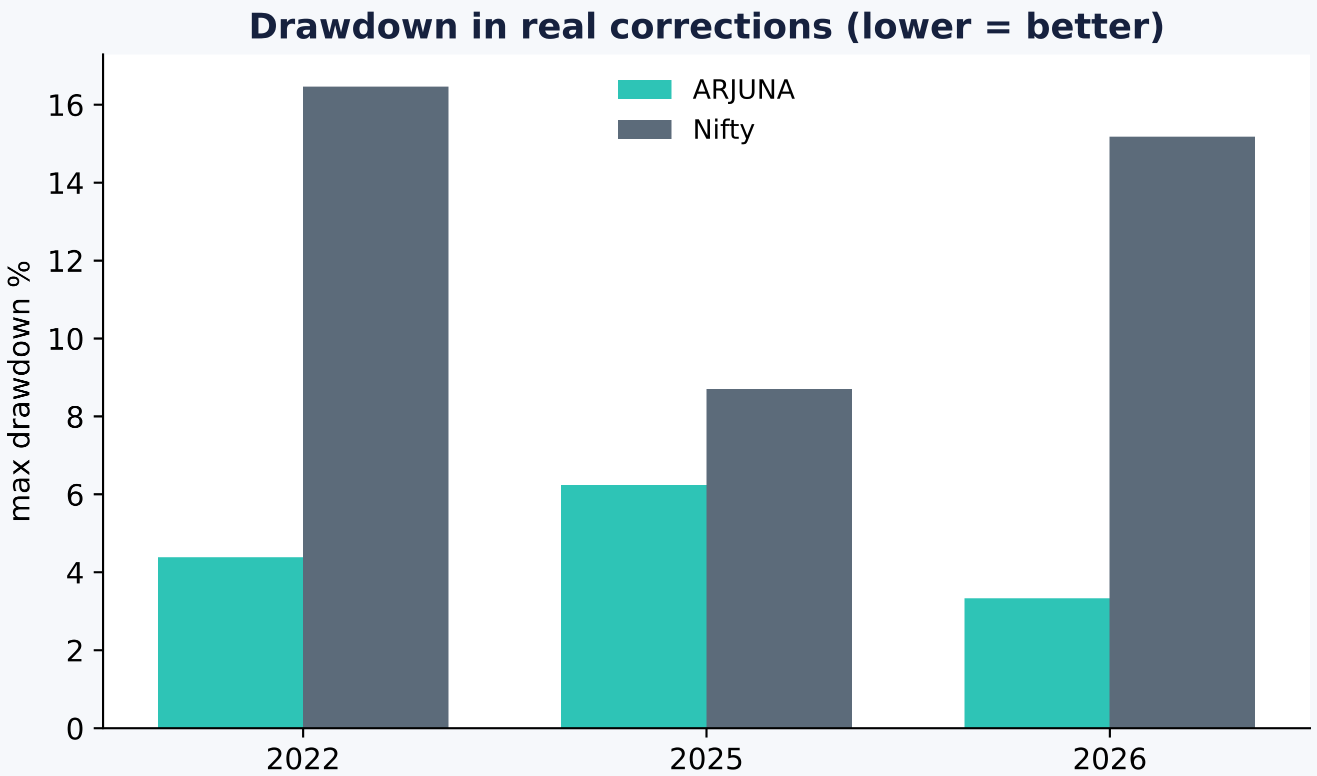
Longer holds → higher probability of profit



≥6 months -> 90%+ positive · ≥1 year -> 96% · ≥2 years -> 100% (in-sample).
ARJUNA is a compounding system — give it time. Suggested horizon: 1 year+.

STRESS & ROBUSTNESS

Built to survive, not just to score



Rolling 3-yr windows: Sharpe 2.1–2.66 (consistent). Monte-Carlo (35% haircut):
median ~9–10%/yr, P(+ve 1yr)=87% / 3yr=98%, P(drawdown>20%) ~ 0% at every haircut.

In 3 of 4 corrections ARJUNA stayed POSITIVE while the Nifty fell hard;
drawdown was 2–3x smaller every time. The regime + Global overlay is the defense.

HOW TO USE IT

A quarterly research desk

CONFIGURE	india/config.py — capital, risk appetite. One file, no logic edits.
QUARTERLY	python india/monthly_snapshot.py -> reports/YYYY_MM.md (dated record).
GET BASKET	python india/run_arjuna.py --retail -> 15-stock buy list + confidence meter.
SIZE TO CAPITAL	auto-scales ₹50k→3 .. ₹10L→15 positions; sector≤2; min-allocation.
HOLD ~1 YEAR	no daily churn; quarterly rebalance; paper_log.csv tracks it forward.
REVIEW	after 4 quarters: did v2.1 survive reality? Promote or investigate.

Golden rule: keep real cash out until the FORWARD paper run beats the index net of co

HONEST EXPECTATIONS

What it will & won't do

WILL: beat the index on risk-adjusted terms · cut drawdowns · de-risk in stress · diversify · avoid news blow-ups · compound steadily (hold 1yr+).

WON'T: pick the next multibagger (impossible in advance) · predict returns · win every year (streaky: great in trends, flat in chop).

Maturity ~98%. Remaining: forward paper (12 mo) + point-in-time fundamentals (data).

THE ARJUNA DOCTRINE

Markets are mostly efficient.	Regime > stock selection.
Returns are noisy.	Construction > models.
Risk has structure.	Data quality > complexity.
Survival > prediction.	Robustness > backtests.
Diversification > concentration.	Long-term compounding > short-term accuracy.